# SMART COURSE REGISTRATION & ACADEMIC PLANNER SYSTEM

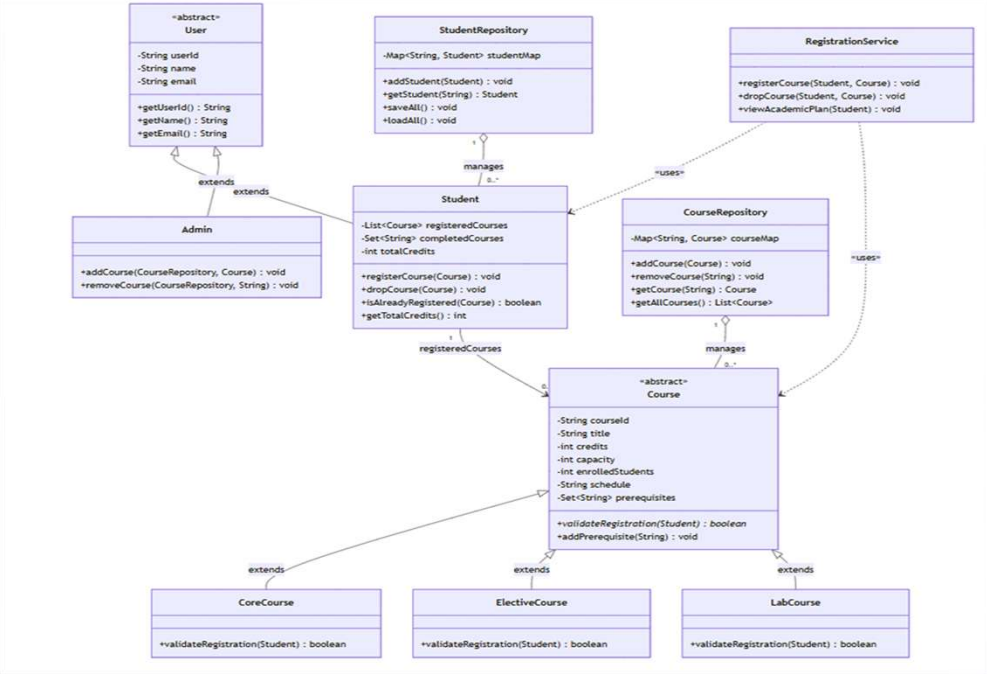
**PROBLEM STATEMENT:**
Traditional systems do not enforce academic constraints like credit limits, prerequisites, timetable clashes, and capacity dynamically.

**TARGET USERS**:
-Undergraduate Students
-Academic Administrators

# SYSTEM DESIGN ARCHITECTURE

## Package Structure

```
courseplanner/
|
├── model/
|   ├── User.java
|   ├── Student.java
|   ├── Admin.java
|   ├── Course.java
|   ├── CoreCourse.java
|   ├── ElectiveCourse.java
|   ├── LabCourse.java
|
├── repository/
|   ├── CourseRepository.java
|   ├── StudentRepository.java
|
├── service/
|   ├── RegistrationService.java
|
├── exception/
|   ├── CreditLimitExceededException.java
|   ├── PrerequisiteNotMetException.java
|   ├── CourseCapacityFullException.java
|   ├── TimeTableConflictException.java
|   ├── DuplicateCourseException.java
|   ├── CourseNotRegisteredException.java
└── ui/
    ├── MainMenu.java
```



# OOP CONCEPTS USED

- **Abstraction**: User and Course abstract classes
- **Inheritance**: Student, Admin → User, Course types → Course
- **Polymorphism**: validateRegistration() method overridden
- **Encapsulation**: Student manages course list internally
- **Design Approach**: Layered Architecture for separation of concerns

# TECHNICAL HIGHLIGHTS

- **Core Logic: Registration Validation**

The system validates multiple constraints before allowing registration:

- Credit limit check
- Prerequisite validation
- Timetable conflict detection
- Course capacity check

**Implementation**:

- Handled in RegistrationService
- Uses custom exceptions:

CreditLimitExceededException

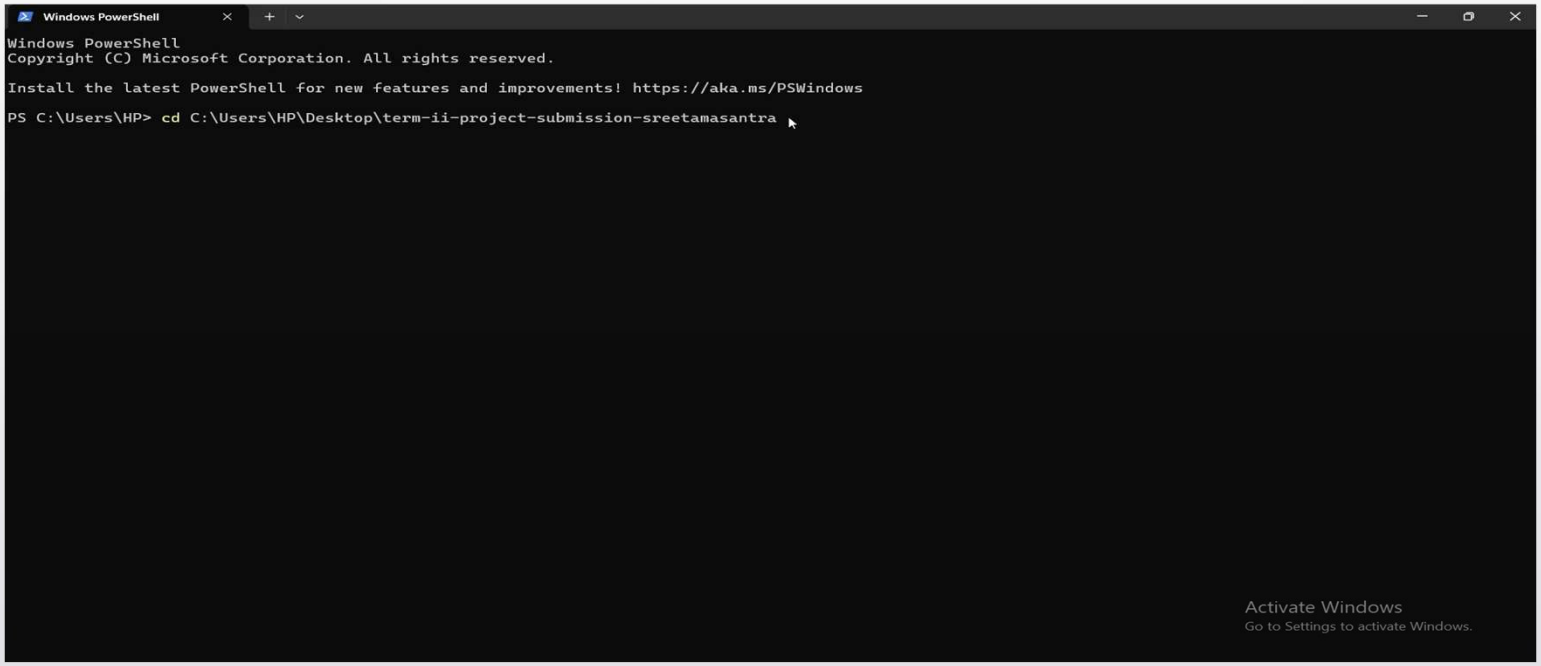
PrerequisiteNotMetException

TimeTableConflictException

CourseCapacityFullException

DuplicateCourseException

# LIVE DEMONSTRATION



THANK YOU !